Communication-Efficient Reinforcement Learning for Linear Quadratic Differential Zero-Sum Games

Tianhao Fei, Zhenzhuo Shan, Yu Wang, Yongliang Yang*, Jianquan Wang
School of Automation and Electrical Engineering
University of Science and Technology Beijing
Beijing, China
feith@ieee.org; shanzhenzhuo@ieee.org;
wangyusally@126.com;
yangyongliang@ieee.org; wangjianquan@ustb.edu.cn

***Abstract*—This paper studies the online Nash solution of continuous-time (CT) linear quadratic zero-sum games (ZSGs) arising from H_∞ control. An actor–critic–disturbance reinforcement learning (ACDRL) algorithm is developed to avoid solving the game algebraic Riccati equation (GARE) offline by directly approximating the value function and Nash equilibrium. To ensure safe learning, projection operators are embedded in the ACDRL to keep the policy and disturbance gains within prescribed compact sets. A novel self-triggered sampling rule is designed to compute the next triggering instant, which removes continuous state monitoring and improves communication efficiency. Rigorous analysis guarantees exponential stability, convergence of the learned strategies, and Zeno-free executions. Simulations validate fast regulation and sparse triggering.**

***Index Terms*—Reinforcement learning, zero-sum game, game algebraic Riccati equation, self-triggered sampling.**

I. Introduction

Zero-sum differential games model antagonistic interactions between a controller and an adversarial disturbance [1], [2]. For nonlinear systems, the Nash equilibrium is characterized by the HJI equation (reducing to a GARE in the linear case) [3], [4]. Yet Hamilton–Jacobi–Isaacs (HJI) is generally intractable and GARE solving can be computationally costly [5]. Adaptive dynamic programming (ADP) circumvents this by approximating the value function and equilibrium policy directly, enabling real-time H_∞ regulation for ZSGs [6]–[9].

In networked and resource-constrained systems, continuous sensing and periodic transmissions are often impractical [10]. Event-triggered control addresses this issue by updating the input only when a state-dependent condition is violated [11]. Combining event-triggering with ADP can reduce communication while preserving stability in game-theoretic control problems [12]. This integration enables online learning of Nash feedback with fewer updates than time-driven sampling [12]. Yet, event-triggering has an intrinsic limitation. The triggering condition must be checked online at all times. This requirement enforces continuous state monitoring and frequent computations. As a result, the communication and processing burden can remain high.

This paper is supported by the National Natural Science Foundation of China under Grant 62373039.
*Corresponding author: Yongliang Yang.

Self-triggered control removes the need for continuous trigger evaluation [13]. Instead of persistently monitoring a triggering inequality, it predicts when the next update will be necessary by exploiting the plant model, the latest sampled state, and pre-specified bounds on disturbances and uncertainties, so the controller can proactively schedule the next transmission and computation. It computes the next update instant at the current sampling time using available state information and known bounds [14]. As a result, the inter-execution interval is explicitly determined and guaranteed to be strictly positive, which eliminates excessive sampling and avoids the pathological accumulation of triggering times in practice. The triggering schedule is therefore generated in an open-loop manner between two transmissions, which significantly lowers sensing effort, communication load, and onboard decision-making frequency. This feature is well aligned with ZSG settings, where both stability and H_∞ control performance must be maintained under limited communication [5].

Accordingly, this work proposes a self-triggered ACDRL framework for the ZSGs, with the following contributions.

1) Conventional event-triggered ADP approaches require continuous trigger monitoring, which can be a burden in systems with limited communication resource [12]. This paper proposes a novel self-triggered scheme that computes the next execution time, which eliminates the intrinsic shortcoming in event-triggering setting.
2) Existing methods do not explicitly enforce the constraints on the adaptive estimations of the players [5], which may lead to instability during the learning process. This paper embeds projection operators into both the actor and disturbance update. This provides stability guarantees while maintaining exponential convergence.

The subsequent sections are arranged as follows. Section II states the ZSG formulation with theoretical preliminaries of the H_∞ regulation problem. The projected ACDRL framework is presented in Section III. The event/self-triggering mechanisms are investigated in Section IV with rigorous theoretical discussions. Simulation studies on a F16 aircraft system are implemented in Section V. Conclusion follows in Section VI.

II. Problem Statement

Consider the following CT linear system

$$\begin{aligned}\dot{x}(t) &= Ax(t) + B_u u(t) + B_d d(t), \quad x(0) = x_0, \\ z(t) &= [C_1 x(t), C_2 u(t)]^\top,\end{aligned} \tag{1}$$

where $x(t) \in \mathbb{R}^n$, $u(t) \in \mathbb{R}$, $d(t) \in \mathbb{R}$, and $z(t)$ are the state, control input, disturbance input, and objective output respectively. A, B_u, B_d, C_1 and C_2 are known constant matrices with appropriate dimensions such that (A, B_u) is stabilizable and (C_1, A) is detectable [15].

Given the instantaneous cost:

$$\mathbf{c}(x, u, d) \triangleq \|z\|^2 - \gamma^2 \|d\|^2, \quad \gamma > 0 \tag{2}$$

the performance associated with (1) is determined as

$$\mathbf{J}(u, d; x_0) \triangleq \int_0^\infty \mathbf{c}(x(t), u(t), d(t)) \mathrm{d}t. \tag{3}$$

For the disturbance input $d(t)$, the H_∞ control problem [15] is to find a stabilizing feedback policy u, such that the system (1) is asymptotically stable at the origin and the following attenuation condition is satisfied:

$$\frac{\int_0^\infty \|z(t)\|^2 \mathrm{d}t}{\int_0^\infty \|d(t)\|^2 \mathrm{d}t} \le \gamma^2, \quad \forall d(t) \in \mathcal{L}_2[0, \infty), \tag{4}$$

where γ is the predetermined disturbance attenuation level [16].

For feedback control policy $u(x)$ and disturbance policy $d(x)$, the value function of the system (1) is defined as

$$\mathbf{V}(x; u, d) \triangleq \int_t^\infty \mathbf{c}\big(x(\tau), u(x(\tau)), d(x(\tau))\big) \mathrm{d}\tau. \tag{5}$$

The value function (5) relates the H_∞ control (i.e., disturbance attenuation) problem (4) to the two-player zero-sum differential game [1], which is specified as

$$\min_u \max_d \mathbf{V}(x; u, d) \tag{6}$$

subject to the system dynamics (1) with minimizing player u and maximizing player d. Then, we aim to find the *Nash equilibrium* (i.e., the saddle point) as defined below.

Definition 1. *[1] The input pair $\{u^*, d^*\}$ is said to constitute a Nash Equilibrium if and only if the following holds*

$$\mathbf{V}(x; u^*, d) \le \mathbf{V}(x; u^*, d^*) \le \mathbf{V}(x; u, d^*), \quad \forall \{u, d\}. \tag{7}$$

According to the dynamic game theory [1], [3], the ZSG (6) admits a unique NE if and only if the saddle point condition [1] holds:

$$\min_u \max_d \mathbf{V}(x; u, d) = \max_d \min_u \mathbf{V}(x; u, d) = \mathbf{V}(x; u^*, d^*),$$

where we define $\mathbf{V}^*(x) \triangleq \mathbf{V}(x; u^*, d^*)$ as the *outcome* of the ZSG (6). Then, to solve the ZSG (6), define the Hamiltonian

$$\mathcal{H}(x, u, d, \nabla \mathbf{V}) = \nabla^\top \mathbf{V}(Ax + B_u u + B_d d) + \mathbf{c}(x, u, d). \tag{8}$$

Enforcing the stationary condition:

$$\frac{\partial \mathcal{H}(x, u, d, \nabla \mathbf{V}^*)}{\partial u} = 0, \quad \frac{\partial \mathcal{H}(x, u, d, \nabla \mathbf{V}^*)}{\partial d} = 0 \tag{9}$$

the Nash solution to the ZSG (6) is given by $u^*(x) = -\frac{1}{2} R^{-1} B_u^\top \nabla \mathbf{V}^*(x)$ and $d^*(x) = \frac{1}{2\gamma^2} B_d^\top \nabla \mathbf{V}^*(x)$, where $R = C_2^\top C_2$. Then, the ZSG (6) is related to the HJI equation

$$\begin{aligned} 0 &= \mathcal{H}(x, u^*, d^*, \nabla \mathbf{V}^*) \\ &= x^\top Q x + \nabla^\top \mathbf{V}^* A x - \frac{1}{4} \nabla^\top \mathbf{V}^* B_u R^{-1} B_u^\top \nabla \mathbf{V}^* \\ &\quad + \frac{1}{4\gamma^2} \nabla^\top \mathbf{V}^* B_d B_d^\top \nabla \mathbf{V}^* \end{aligned} \tag{10}$$

where $Q = C_1^\top C_1$.

Since the system (1) is linear with the quadratic cost (2), the value function $\mathbf{V}^*(x)$ can be represented in a quadratic form [4], [7], i.e., $\mathbf{V}^*(x) = x^\top P x$ with $P = P^\top \succ 0$. Thus, the Nash Equilibrium $\{u^*, d^*\}$ can be rewritten as

$$\begin{aligned} u^*(x) &= K_u^* x, \quad K_u^* = -R^{-1} B_u^\top P, \\ d^*(x) &= K_d^* x, \quad K_d^* = \frac{1}{\gamma^2} B_d^\top P. \end{aligned} \tag{11}$$

Inserting the Nash equilibrium $\{u^*, d^*\}$ in (11) into the HJI equation (10) yields the GARE [2]:

$$A^\top P + PA + Q - P B_u R^{-1} B_u^\top P + \frac{1}{\gamma^2} P B_d B_d^\top P = 0. \tag{12}$$

Then, to find the Nash solution (11) to the ZSG (6) is to find the positive definite solution to the GARE (12).

III. Online Solution to the GARE

A. Sampling Process

To save communication resource in the dynamical environment, we implement the sampling procedure on the system state $x(t)$ such that

$$\check{x}(t) = x(t_i), \quad t \in [t_i, t_{i+1}), \tag{13}$$

with an increasing time sequence $\langle t_i \rangle_{i=0}^\infty$, the sampled state $\check{x}$, and the sampling error $\tilde{x} = x - \check{x}$.

Then, the input pair with intermittent/continuous feedback can be rewritten as

$$u^*(\check{x}) = K_u^* \check{x}, \quad d^*(x) = K_d^* x. \tag{14}$$

B. Representation of the value function and Nash Equilibrium

Rather than computing the GARE solution in an iterative way, which brings computational burden, we directly estimate the Nash solution P and the Nash equilibrium gains K_u^*, K_d^*.

Specifically, the estimations of P, K_u^*, and K_d^* are denoted as $\hat{P}$, $\hat{K}_u$, and $\hat{K}_d$, respectively. Then, the value function $\mathbf{V}^*$ and the Nash equilibrium $\{u^*, d^*\}$ can be approximated as

$$\hat{\mathbf{V}}(t) = P_c^\top(t) \phi(x(t)), \tag{15}$$

$$\hat{u}_1(t) = \hat{K}_u(t) \check{x}(t), \tag{16}$$

$$\hat{d}_1(t) = \hat{K}_d(t) x(t), \tag{17}$$

where $P_c = \mathrm{vech}(\hat{P}) \in \mathbb{R}^{n(n+1)/2}$ is the vector obtained by stacking columns of the symmetric matrix $\hat{P}$ on top of one another. The subscript "1" indicates the behavior policy implemented on the system (1). $\phi(x) \in \mathbb{R}^{n(n+1)/2}$ is the polynomial basis function of $x = [x_1, \cdots, x_n]^\top$, i.e.,

$$\phi(x) = [x_1^2, x_1 x_2, \cdots, x_1 x_n, x_2^2, \cdots, x_2 x_n, \cdots, x_n^2]^\top \tag{18}$$

C. ACDRL for the Nash Equilibrium

In this subsection, we present the ACDRL algorithm to obtain the approximate Nash equilibrium solution $\{\hat{u}, \hat{d}\}$ to the ZSG (6).

*1) **Critic Learning**:* Substituting $\{\hat{\mathbf{V}}, \hat{u}, \hat{d}\}$ in (15), (16), and (17) into the Hamiltonian (8) yields

$$\mathcal{H}(x, \hat{u}_1(t), \hat{d}_1(t), \nabla\hat{\mathbf{V}}) = P_c^\top \sigma(t) + \mathbf{c}(x, \hat{u}(t), \hat{d}(t)), \quad (19)$$

where $\sigma \triangleq \nabla\phi(Ax + B_u\hat{u}_1 + B_d\hat{d}_1)$.

Therefore, the critic learning error given by

$$\begin{aligned}&\epsilon_c(t)\\ =&\mathcal{H}(x, \hat{u}_1(t), \hat{d}_1(t), \nabla\hat{\mathbf{V}}(t)) - \mathcal{H}(x, u^*(x), d^*(x), \nabla\mathbf{V}^*(x))\\ =&P_c^\top\sigma(t) + \mathbf{c}(x, \hat{u}_1(t), \hat{d}_1(t)).\end{aligned}$$

Accordingly, the objective function to be minimized by critic learning is designed as

$$\mathbf{E}_c(t) = \frac{\epsilon_c^2(t)}{2[1 + \|\sigma(t)\|^2]^2}.$$

Then, by gradient descent, the update law for critic is

$$\dot{P}_c(t) = -\eta_c\frac{\partial\mathbf{E}_c(t)}{\partial P_c(t)} = -\eta_c\frac{\sigma\epsilon_c(t)}{[1 + \|\sigma(t)\|^2]^2}, \quad (20)$$

with the critic learning rate $\eta_c > 0$.

*2) **Actor Learning**:* Given the critic estimation in (15), the target for actor learning is defined as $\hat{u}_2(t) = -\frac{1}{2}R^{-1}B_u^\top\nabla^\top\phi(\check{x}(t))P_c(t)$. Combined with (16), the actor learning error is established as $\epsilon_u(t) = \hat{u}_1(t) - \hat{u}_2(t)$. Then, the objective function for actor learning is $\mathbf{E}_u(t) = \frac{1}{2}\epsilon_u^2(t)$.

To exclude divergence during the learning process, we introduce the *projection operator* for the actor, as follows.

Definition 2. *[17] Given a vector $\vartheta \in \mathbb{R}^p$ and a constant $\beta > 0$, the discrete projection operator is*

$$\mathbf{Proj}_u(\vartheta|\beta) = \vartheta + \mathbf{1}\{\beta < \|\vartheta\|\} \cdot \frac{\beta - \|\vartheta\|}{\|\vartheta\|}\vartheta.$$

Then, the projected actor tuning law is designed as

$$\hat{K}_u^+(t) = \mathbf{Proj}_u\big(\bar{K}_u(t)|\rho_u\big), \quad t = t_{i+1}, \quad (21)$$

where $\bar{K}_u \triangleq \hat{K}_u - \eta_u\check{x}^\top\epsilon_u$, $\eta_u > 0$ is the actor learning rate, and ρ_u is a predetermined positive constant. Thus, the actor update in (21) has following property

Lemma 1. *[17] Under the actor update law* (21) *with proper η_u, the actor weight $\hat{K}_u$ is bounded satisfying $\|\hat{K}_u(t)\| \leq \rho_u$, for all $t \geq 0$, provided that $\|\hat{K}_u(0)\| \leq \rho_u$ and $\|K_u^*\| \leq \rho_u$.*

3) Disturbance Learning: Based on the approximated value function in (15), the disturbance input is to approximate $\hat{d}_2(t) = \frac{1}{2\gamma^2}B_d^\top\nabla^\top\phi(x(t))P_c(t)$. Similar to the actor learning, the disturbance learning error is $\epsilon_d(t) = \hat{d}_1(t) - \hat{d}_2(t)$. The disturbance learning aims to minimize $\mathbf{E}_d(t) = \frac{1}{2}\epsilon_d^2(t)$.

To guarantee the estimation stability during the ACD learning, the projection operator for the disturbance is implemented.

Definition 3. *[17] Given vectors $\vartheta, \omega \in \mathbb{R}^p$, a positive definite matrix $\Phi \in \mathbb{R}^{p\times p}$, a predefined convex set Ω_ϑ, and a constant $\beta > 0$, the continuous projection operator is defined as*

$$\mathbf{Proj}_d\big(\omega, \vartheta\big|\beta, \Omega_\vartheta, \Phi\big) = \omega - \mathbf{1}\{\vartheta \in \Omega_\vartheta\} \cdot \Phi \cdot \frac{\vartheta\vartheta^\top}{\|\vartheta\|_\Phi^2} \cdot \omega,$$

with $\Omega_\vartheta \triangleq \{\vartheta\big|\|\vartheta\| = \beta \text{ and } \omega^\top\vartheta > 0\}$.

Enforcing above projection operator, the disturbance learning law is formulated as

$$\dot{\hat{K}}_d(t) = \mathbf{Proj}_d\big(-\eta_d x^\top\epsilon_d, \hat{K}_d(t)\big|\rho_d, \Omega_d, \eta\big) \quad (22)$$

with user-defined constraint $\rho_d > 0$, learning rate $\eta_d > 0$, and $\Omega_d \triangleq \{\hat{K}_d\big|\|\hat{K}_d\| = \rho_d \cap (-\eta_d x^\top\epsilon_d)^\top\hat{K}_d > 0\}$. This leads to the following lemma.

Lemma 2. *[17] Let the disturbance gain evolves as* (22) *with $\hat{K}_d(0) \in \Omega_d$, $K_d^* \in \Omega_d$, and proper learning rate $\eta_d > 0$. Then, $\|\hat{K}_d(t)\| \leq \rho_d$ for all $t \geq 0$.*

IV. Triggering Mechanisms

In what follows, we investigate the sampling rules for the state to improve communication efficiency while guaranteeing stability of the dynamical system (1).

A. Event-triggered sampling

To begin with, we introduce the commonly adopted event-triggering mechanism, with the triggering condition:

$$\|\tilde{x}(t)\|^2 \leq \delta_1\|x(t)\|^2 + \delta_2, \quad (23)$$

where δ_1 and δ_2 are positive parameters to be designed.

Following standard assumption are required.

Assumption 1. *[10] At any sample instant, the finite-time stabilization is not achieved, i.e., $\|\check{x}(t_i)\| \neq 0$, for all $i \in \mathbb{N}$.*

The properties of event-driven ACDRL can be summarized in the following algorithm.

Algorithm 1 Event-triggered mechanism for the ZSG

1: **Initialization:**
2: Set parameters δ_1 and δ_2 in the condition (23);
3: Set $i \leftarrow 0$, initial time $t_i \leftarrow 0$ and running time t_{span};
4: Set initial states $x(t_i)$ randomly and $\check{x}(t_i) \leftarrow x(t_i)$;
5: Set initial estimation $P_c(0)$, $\hat{K}_u(0)$, and $\hat{K}_d(0)$; set constraints ρ_u and ρ_d; set learning rates η_c, η_u, and η_d.
6: **while** $t < t_{\text{span}}$ **do**
7: Propagate t and $x(t)$ by (1), (16), (17), and the ACDRL laws (20) and (22) by the ordinary differential equation solver with the event-triggering condition (23);
8: Update the sampled state: $\check{x} \leftarrow x(t(\text{end}))$;
9: Update the actor gain $\hat{K}_u \leftarrow \hat{K}_u^+$ by (21);
10: Update the actor: $\hat{u}_1 = \hat{K}_u\check{x}$;
11: Update the trigger instant: $t_i \leftarrow t(\text{end})$.
12: **end while**

The event-based ACDRL design in Algorithm 1 can be summarized in the following theorem.

Theorem 1. *Under Assumption 1 and sampling procedure* (13)*, consider the event triggering condition* (23) *with*

$$\delta_1 = \frac{\kappa\lambda_{\min}(Q)}{18\lambda_{\max}^2(R)\rho_u^2}, \quad \delta_2 = \frac{\|\hat{u}_1\|_R^2}{18\lambda_{\max}^2(R)\rho_u^2}, \tag{24}$$

where $\kappa \in (0,1)$. *Let the value function and Nash equilibrium be approximated as* (15)-(17) *with the ACDRL algorithm in Section III-C. Then, the following statements hold:*

1) *the state* $x(t)$ *converges to the origin exponentially;*
2) *the pair* $\{\hat{u}_1, \hat{d}_1\}$ *converges to the Nash equilibrium* $\{u^*, d^*\}$ *of the ZSG* (6)*;*
3) *the event triggering mechanism* (23) *is Zeno-free.*

Proof. The theorem will be proven item by item.

1) First, we select $\mathbf{V}^*(x)$ as the Lyapunov candidate, whose time derivative along (1) is

$$\dot{\mathbf{V}}^*(x) = 2x^\top P(Ax + B_u\hat{u}_1 + B_d\hat{d}_1). \tag{25}$$

By the GARE (12), the term $2x^\top PAx$ satisfies

$$2x^\top PAx = \|u^*\|_R^2 - \gamma^2\|d^*\|^2 - \|x\|_Q^2.$$

With the facts $2x^\top PB_u\hat{u}_1 = -2(u^*)^\top R\hat{u}_1$ and $2x^\top PB_d\hat{d}_1 = 2\gamma^2(d^*)^\top\hat{d}_1$, $\dot{\mathbf{V}}^*$ in (25) can be rewritten as

$$\begin{aligned}&\dot{\mathbf{V}}^*\\ =& -\|x\|_Q^2 + \|u^* - \hat{u}_1\|_R^2 - \gamma^2\|d^* - \hat{d}_1\|^2 - \|\hat{u}_1\|_R^2 + \gamma^2\|\hat{d}_1\|^2.\end{aligned}$$

For the term $\|u^* - \hat{u}_1\|_R^2$, one has

$$\begin{aligned}&\|u^* - \hat{u}_1\|_R^2 = \|u^*(x) - u^*(\check{x}) + u^*(\check{x}) - \hat{u}_1(t)\|_R^2\\ \le& 2\lambda_{\max}^2(R)\|K_u^*\|\|\tilde{x}\|^2 + 2\lambda_{\max}^2(R)\big\|K_u^* - \hat{K}_u\big\|^2\|\check{x}\|^2.\end{aligned}$$

Moreover, by the projection actor tuning law in (21) with the property in Lemma 1, one has $\|K_u^* - \hat{K}_u\|^2\|\check{x}\|^2 = \|K_u^* - \hat{K}_u\|^2\|x - \tilde{x}\|^2 \le (2\|K_u^*\|^2 + 2\|\hat{K}_u\|^2)(2\|x\|^2 + 2\|\tilde{x}\|^2) \le 8\rho_u^2(\|x\|^2 + \|\tilde{x}\|^2)$. Then, the following inequality holds

$$\|u^* - \hat{u}_1\|_R^2 \le 16\lambda_{\max}^2(R)\rho_u^2\|x\|^2 + 18\lambda_{\max}^2(R)\rho_u^2\|\tilde{x}\|^2 \tag{26}$$

Considering $-\|x\|_Q^2 \le -\lambda_{\min}(Q)\|x\|^2$ and $-\gamma^2\|d^* - \hat{d}_1\|^2 \le 0$, the derivative $\dot{\mathbf{V}}^*$ can be further expressed as

$$\begin{aligned}\dot{\mathbf{V}}^* \le& -\big[\underbrace{\bar{\kappa}\lambda_{\min}(Q) - 16\lambda_{\max}^2(R)\rho_u^2 - \gamma^2\rho_d^2}_{\triangleq\varrho}\big]\|x\|^2\\ &-\kappa\lambda_{\min}(Q)\|x\|^2 + 18\lambda_{\max}^2\rho_u^2\|\tilde{x}\|^2 - \|\hat{u}_1\|_R^2,\end{aligned} \tag{27}$$

According to the event-triggering condition (23) and parameters design in (24), one has

$$-\kappa\lambda_{\min}(Q)\|x\|^2 + 18\lambda_{\max}^2\rho_u^2\|\tilde{x}\|^2 - \|\hat{u}_1\|_R^2 \le 0 \tag{28}$$

over each execution intervals. Inserting (28) into (27) yields

$$\dot{\mathbf{V}}^* \le -\varrho\|x\|^2. \tag{29}$$

By selecting the parameters κ, Q, R, γ, ρ_u, and ρ_d, one can guarantee $\varrho > 0$. By the comparison Lemma in [18], one has

$$\|x(t)\| \le x_0\sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}}\cdot \mathrm{e}^{-\frac{\varrho t}{\lambda_{\max}(P)}}, \quad \forall t \ge 0, \tag{30}$$

which verifies the first conclusion.

2) Based on (11), (16), and (17), one has

$$\begin{aligned}u_{\mathrm{error}}(t) &\triangleq \|u^*(x(t)) - \hat{u}_1(t)\| \le \rho_u(\|\tilde{x}(t)\| + 2\|\check{x}(t)\|),\\ d_{\mathrm{error}}(t) &\triangleq \|d^*(x(t)) - \hat{d}_1(t)\| \le \rho_d\|x(t)\|.\end{aligned}$$

Since $x(t)$ converges to the origin exponentially, i.e., $x(t) \to 0$ as $t \to \infty$ and thus $\tilde{x}(t) \to 0$ and $\check{x}(t) \to 0$, one has

$$u_{\mathrm{error}}(t) \to 0, \quad d_{\mathrm{error}}(t) \to 0 \quad \text{as} \quad t \to \infty. \tag{31}$$

This implies $\{\hat{u}_1, \hat{d}_1\}$ converges to the Nash equilibrium.

3) The third property is referred to [10, Theorem III]. This completes the proof. □

B. Self-triggered sampling

It should be noted that in the event triggering setting, the state $x(t)$ need to be monitored over the entire time interval to check if the condition (23) is satisfied. This can be a waste of communication resource and brings computational overhead.

In what follows, a novel self-triggered sampling mechanism will be developed to eliminate this difficulty, which requires a standard assumption.

Assumption 2. *[13] For all stabilizable* u *and disturbance* $d \in \mathcal{L}_2$, $\exists\, \xi > 0$ *such that* $\forall\, x, \check{x}, \tilde{x} \in \mathbb{R}^n$, *the following holds:*

$$\begin{aligned}x^\top(Ax + B_uu + B_dd) &\le \xi(1 + \|x\|^2 + \|\tilde{x}\|^2),\\ \tilde{x}^\top(Ax + B_uu + B_dd) &\le \xi(1 + \|x\|^2 + \|\tilde{x}\|^2).\end{aligned}$$

Before presenting the design of self-triggered sampling rule, we give the following lemma that relates the sampled state $\check{x}(t)$ to the sampling error $\tilde{x}(t)$ and system state $x(t)$.

Lemma 3. *[13] Let Assumption 2 hold. Then, for all* $t \ge 0$*:*

$$\|\tilde{x}(t)\|^2 \le \Xi_1(\|\check{x}(t)\|, t - t_i), \tag{32}$$
$$\|x(t)\|^2 \ge \Xi_2(\|\check{x}(t)\|, t - t_i), \tag{33}$$

with $\Xi_1(\|\check{x}\|, t - t_i) \triangleq \frac{2\|\check{x}\|^2+1}{3}(\mathrm{e}^{6\xi t} - 1)$ *and* $\Xi_2(\|\check{x}\|, t - t_i) = \frac{5\|\check{x}\|^2+1}{3}\mathrm{e}^{-6\xi t} - \frac{2\|\check{x}\|^2+1}{3}$.

Then, we are ready to state the main results of this paper.

***Self-triggered mechanism**:* Consider the sampling rule

$$\Psi_2\big(\Xi_1(\|\check{x}\|, t - t_i)\big) \le \omega_1\Psi_1\big(\Xi_2(\|\check{x}\|, t - t_i) + \omega_2\big), \tag{34}$$

where the parameters ω_1, ω_2 and functions Ψ_1, Ψ_2 satisfy:

1) $\omega_1, \omega_2 > 0$;
2) $\Psi_1 \in \mathcal{K}$, $\Psi_2 \in \mathcal{K}_\infty$ and $\Phi_1(\chi) \le \chi \le \Psi_2(\chi)$, $\forall\, \chi \ge 0$.

Then, the following inequality can be derived

$$\begin{aligned}\|\tilde{x}\|^2 &\le \Psi_2(\|\tilde{x}\|^2) \le \Psi_2\big(\Xi_1(\|\check{x}\|, t - t_i)\big)\\ &\le \omega_1\Psi_1\big(\xi_2(\|\check{x}\|, t - t_i) + \omega_2\big)\\ &\le \omega_1\Psi_1\big(\|x\|^2 + \omega_2\big) \le \omega_1\|x\|^2 + \omega_1\omega_2.\end{aligned} \tag{35}$$

From (35), one can observe that the self-triggering condition (34) is more conservative than the event-triggering condition (23). That is, (34) is a sufficient condition to (23). Furthermore,

(34) is more computationally intelligent than (23), since the successive triggering instants can be determined by solving

$$\Psi_2\big(\Xi_1(\|\check{x}\|, t - t_i)\big) = \omega_1 \Psi_1\big(\Xi_2(\|\check{x}\|, t - t_i) + \omega_2\big) \quad (36)$$

when ω_1, ω_2, Ψ_1, and Ψ_2 are defined.

Collecting the discussions set forth the self-triggered ACDRL algorithm can be summarized in the following pseudocode. Given the current triggering instant t_i, the successive sample time t_{i+1} can be computed by solving (36), and the system (1) evolves with $\{\hat{u}_1, \hat{d}_1\}$.

Algorithm 2 Self-triggered mechanism for the ZSG

Initialization:
Select parameters ξ, ω_1 and ω_2 in the condition (34);
Set $i \leftarrow 0$, initial time $t_i \leftarrow 0$ and running time t_{span};
Set initial states $x(t_i)$ randomly and $\check{x}(t_i) \leftarrow x(t_i)$;
Set initial estimation $P_c(0)$, $\hat{K}_u(0)$, and $\hat{K}_d(0)$; set constraints ρ_u and ρ_d; set learning rates η_c, η_u, and η_d.
while $t < t_{\text{span}}$ **do**
  Compute the triggering instant t_{i+1} by (36);
  Propagate t and $x(t)$ by (1), (16), (17), and the ACDRL laws (20) and (22) by the ordinary differential equation solver on the interval $[t_i, t_{i+1}]$;
  Update the sampled state: $\check{x}(t) \leftarrow x(t_{i+1})$;
  Update the actor gain $\hat{K}_u \leftarrow \hat{K}_u^+$ by (21);
  Update the actor: $\hat{u}_1 \leftarrow \hat{K}_u \check{x}$;
  Update the trigger instant: $t_i \leftarrow t_{i+1}$.
end while

The properties of the self-triggered ACDRL in Algorithm 2 can be summarized in the following theorem.

Theorem 2. *Under Assumption 2 and the sampling procedure* (13)*, consider the self-triggered sampling rule* (34) *with*

$$\omega_1 = \frac{\kappa \lambda_{\min}(Q)}{18\lambda_{\max}^2(R)\rho_u^2}, \quad \omega_2 = \frac{\|\hat{u}_1\|_R^2}{\kappa \lambda_{\min}(Q)}. \quad (37)$$

where $\kappa \in (0,1)$. *Let the value function and Nash equilibrium be approximated as* (15)-(17) *with the ACDRL algorithm in Section III-C. Then, the following statements hold:*

1) *the state* $x(t)$ *converges to the origin exponentially;*
2) *the pair* $\{\hat{u}_1, \hat{d}_1\}$ *converges to the Nash equilibrium* $\{u^*, d^*\}$ *of the ZSG* (6)*;*
3) *the self-triggering mechanism* (34) *is Zeno-free.*

Proof. The theorem will be proven item by item.

1) Based on the discussions in the event-triggered setting (see (25)-(27)), one has

$$\dot{\mathbf{V}}^* \leq -\varrho\|x\|^2 - \kappa\lambda_{\min}(Q)\|x\|^2 + 18\lambda_{\max}^2\rho_u^2\|\tilde{x}\|^2 - \|\hat{u}_1\|_R^2,$$

where $\varrho \triangleq \bar{\kappa}\lambda_{\min}(Q) - 16\lambda_{\max}^2(R)\rho_u^2 - \gamma^2\rho_d^2$ for $\bar{\kappa} = 1 - \kappa$. Then, with the inequality (35) and the parameters in (37), one has $-\kappa\lambda_{\min}(Q)\|x\|^2 + 18\lambda_{\max}^2\rho_u^2\|\tilde{x}\|^2 - \|\hat{u}_1\|_R^2 \leq 0$, which implies $\dot{\mathbf{V}}^* \leq -\varrho\|x\|^2$. By the comparison Lemma in [18], one has $\|x(t)\| \leq x_0\sqrt{\lambda_{\max}(P)/\lambda_{\min}(P)} \cdot \mathrm{e}^{-\frac{\varrho t}{\lambda_{\max}(P)}}, \forall t \geq 0$, which verifies the first conclusion.

2) The second conclusion is identical to previous discussions in Theorem 1.

3) Since the functions Ξ_1 and Ξ_2 are monotonically decreasing and increasing in t, the successive triggering instant t_{h+1} can be uniquely determined by explicitly solving (36), i.e., the triggering instant t_{i+1} uniquely exists for all $i \in \mathbb{N}$.

Next, we prove the Zeno-free property. Consider a function $\phi(\cdot)$ such that $0 < \phi(\|\check{x}\|) < \|\check{x}\|^2$. Let the $\phi(\cdot)$ further satisfy $\Xi_2(\|\check{x}\|, \Delta t_i) + \omega_2 \leq \phi(\|\check{x}\|) \leq \Xi_1(\|\check{x}\|, \Delta t_i)$, which yields

$$\Delta t_i \geq \frac{1}{6\xi}\min\Big\{\ln\Big(1 + \frac{3\phi(\|\check{x}\|)}{1 + 2\|\check{x}\|^2}\Big), \\ -\ln\Big(1 - \frac{3(\omega_2 + \|\check{x}\|^2 - \phi(\|\check{x}\|))}{1 + 5\|\check{x}\|^2}\Big)\Big\} > 0$$

This completes the proof. $\square$

V. Simulation

The efficacy of the proposed self-triggered ACDRL algorithm will be verified by the perturbed F16 aircaft plant [19]:

$$\dot{x} = \begin{bmatrix} -1.01887 & 0.90506 & -0.00215 \\ 0.82225 & -1.07741 & -0.17555 \\ 0 & 0 & -1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} d,$$

with quadratic cost function such that $Q = 10I_3$, $R = 1$, $\gamma = 5$. The learning rates of the ACDRL algorithm presented in Section III-C are $\eta_c = 0.8$, $\eta_u = 0.1$, and $\eta_d = 0.1$. The self-triggering condition in (34) is determined with $\kappa = 0.5$, $\xi = 0.1$, and $\Psi_1(\chi) = \Psi_2(\chi) = \chi$.

Based on these settings, the trajectory of system state is shown in Figure 1, which converges to the origin exponentially fast, confirming the theoretical analysis in Theorem 2. The evolution of the input pair is depicted in Figure 2, which converges to the Nash equilibrium of the ZSG (6). Figure 3 illustrates the self-triggering intervals, which increases as the system is stabilized. The disturbance attenuation effect is drawn in Figure 4, which converges to 4.23 (below the given level $\gamma = 5$). This indicates that the "worst-case" input disturbance $\hat{d}_1$ is attenuated by the action player $\hat{u}_1$, i.e., the condition (4) of the H_∞ regulating problem is satisfied.

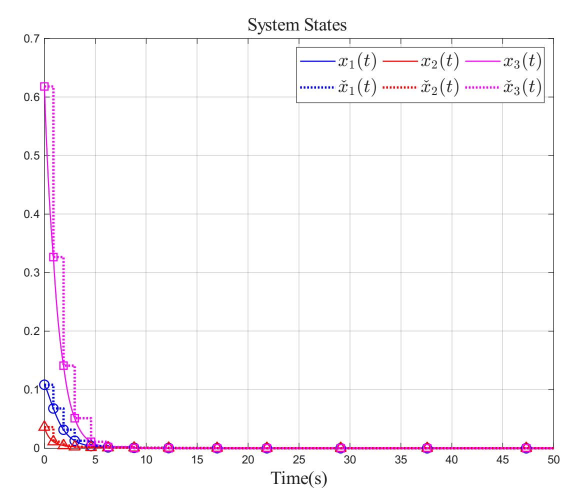

Fig. 1. Evolution of the system states.

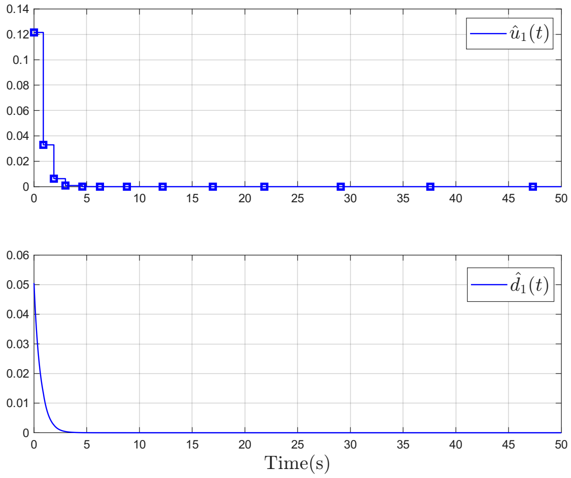

Fig. 2. Evolution of the input pair.

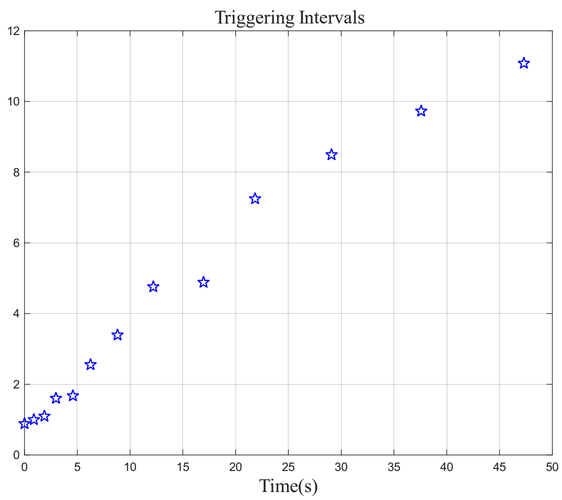

Fig. 3. Execution intervals of the STC mechanism.

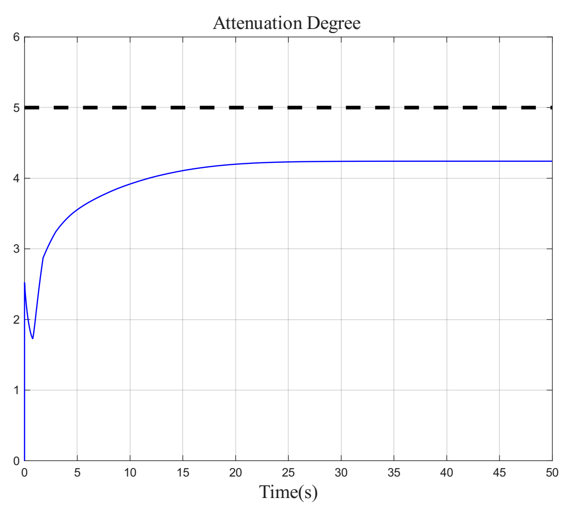

Fig. 4. Disturbance attenuation level of the proposed method.

VI. Conclusion

This paper develops a self-triggered ACDRL framework for CT linear quadratic ZSGs. To compute the GARE solution online, a projected learning scheme is introduced to confine both the actor parameters and the disturbance estimator within prescribed compact sets, thereby enforcing safety-oriented bounds. A self-triggered sampling rule is designed to remove the need for continuous state monitoring and to enhance communication efficiency. Rigorous analysis proves closed-loop stability and establishes Zeno-free behavior. Numerical studies further demonstrate exponential convergence and sparse updates of the self-triggered ACDRL algorithm. Future work will extend the results to multi-agent nonlinear systems.

References

[1] T. Başar and P. Bernhard, *H_∞ optimal control and related minimax design problems: a dynamic game approach*. Boston, MA, USA: Springer, 2008.

[2] T. Başar and G. J. Olsder, *Dynamic noncooperative game theory*. Philadelphia, PA, USA: SIAM, 1999.

[3] J. L. Speyer and D. H. Jacobson, *Primer on optimal control theory*. Philadelphia, PA, USA: SIAM, 2010.

[4] D. Vrabie, K. G. Vamvoudakis, and F. L. Lewis, *Optimal adaptive control and differential games by reinforcement learning principles*. London, UK: Institution of Engineering and Technology, 2012.

[5] Y. Yang, H. Modares, K. G. Vamvoudakis, and F. L. Lewis, "Cooperative finitely excited learning for dynamical games," *IEEE Transactions on Cybernetics*, vol. 54, no. 2, pp. 797–810, 2024.

[6] Y. Yang, W. Gao, and Z.-P. Jiang, "Fast adaptive dynamic programming: Shamanskii iteration for adaptive optimal control," *Automatica*, vol. 181, p. 112494, 2025.

[7] F. L. Lewis, D. Vrabie, and V. L. Syrmos, *Optimal control*. Hoboken, NJ, USA: Wiley, 2012.

[8] Y. Yang, Y. Pan, C.-Z. Xu, and D. C. Wunsch, "Hamiltonian-driven adaptive dynamic programming with efficient experience replay," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 3, pp. 3278–3290, 2024.

[9] Y. Yang, B. Kiumarsi, H. Modares, and C. Xu, "Model-free λ-policy iteration for discrete-time linear quadratic regulation," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 2, pp. 635–649, 2023.

[10] P. Tabuada, "Event-triggered real-time scheduling of stabilizing control tasks," *IEEE Transactions on Automatic Control*, vol. 52, no. 9, pp. 1680–1685, 2007.

[11] A. Girard, "Dynamic triggering mechanisms for event-triggered control," *IEEE Transactions on Automatic Control*, vol. 60, no. 7, pp. 1992–1997, 2015.

[12] W. Xu, T. Wang, J. Qiu, and X. Liu, "A novel framework for game-based optimal event-triggered control of multi-input nonlinear systems," *IEEE Transactions on Automatic Control*, vol. 69, no. 7, pp. 4867–4874, 2024.

[13] R. P. Anderson, D. Milutinović, and D. V. Dimarogonas, "Self-triggered sampling for second-moment stability of state-feedback controlled sde systems," *Automatica*, vol. 54, pp. 8–15, 2015.

[14] Y.-F. Gao, X.-M. Sun, C. Wen, and W. Wang, "Estimation of sampling period for stochastic nonlinear sampled-data systems with emulated controllers," *IEEE Transactions on Automatic Control*, vol. 62, no. 9, pp. 4713–4718, 2016.

[15] K.-Z. Liu and Y. Yao, *Robust control: theory and applications*. Singapore: Wiley, 2016.

[16] A. van der Schaft, *L_2-gain and passivity techniques in nonlinear control*. New York, NY, USA: Springer-Verlag, 1996.

[17] P. A. Ioannou and J. Sun, *Robust Adaptive Control*. Upper Saddle River, NJ, USA: Prentice Hall, 1996.

[18] H. Khalil, *Nonlinear Systems*, 3rd ed. Upper Saddle River, NJ, USA: Prentice Hall, 2002.

[19] B. L. Stevens and F. L. Lewis, *Aircraft control and simulation*, 2nd ed. New Jersey: John Wiley, 2003.